Usman Khalid

Senior Software Engineer | Technical Lead | Distributed Systems & Applied AI

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SUMMARY

Senior Software Engineer and Technical Lead with 9+ years designing scalable, fault-tolerant distributed systems and leading cross-functional teams. Spearheaded a multi-hundred-gigabyte database architecture migration that deprecated MongoDB entirely and cut thousands of dollars a month in cloud costs, overhauled CI/CD and observability to hold a 99% uptime SLA, and built applied-AI tooling that accelerated PR approvals by 35% while halving manual QA time. Leads by mentoring, reviewing and setting technical direction rather than by title.

WORK EXPERIENCE

Senior Full-Stack Engineer / Technical Lead

Feb 2024 - Apr 2026

Premion (TEGNA) | California, USA (Remote)

- Architected and maintained high-scale Node.js microservices (Express, Fastify) powering an industry-leading CTV/OTT advertising platform running high-volume ad campaigns.
- Delivered attribution systems end to end across the Forecast, Creative and Attribution modules, exposing Swagger-documented REST APIs consumed across the platform.
- Designed the real-time analytics storage strategy across MongoDB, PostgreSQL and Snowflake; drove the migration off MongoDB as data grew into hundreds of gigabytes, reducing cost and improving analytical join performance.
- Held system uptime through significant post-acquisition data growth via SQL query optimization, indexing, and AWS and Docker infrastructure tuning.
- Led code review and approval for junior developers; set technical direction on schema and module design, and mentored the team on performance patterns such as eliminating N+1 queries.
- Owned sprint scope and delivery in coordination with the CEO and product manager; tracked work in Jira and versioned in Git.
- Engineered Prism PR Copilot, an open-source AI-driven CI/CD tool using the Anthropic Claude API to validate Jira acceptance criteria against pull requests, accelerating PR approvals by 35% and saving 50% of manual QA validation time.

Senior Full-Stack Engineer

Apr 2022 - Feb 2024

Octillion | California, USA (Remote) — acquired by Premion

- Designed PostgreSQL and Snowflake schemas carrying high-throughput CTV/OTT advertising data at platform scale.
- Built Node.js APIs and migrated the services behind them onto Docker containers on AWS, making deployments consistent across every environment.
- Owned the database layer end to end — schema design, indexing and query optimization — as advertising data volume outgrew the original architecture.

Engineer

May 2020 - May 2022

Sprout Tech | Remote

- Launched Content Gorilla from scratch and scaled it to 20,000+ active users, building and maintaining the full stack on Laravel and Vue.js.
- Built an automated AI content engine that processed 178M+ words and 500k+ blog posts, integrated with the WordPress and Medium REST APIs.

- Automated video-to-blog conversion and social scheduling, and kept content delivery reliable at scale by wiring third-party publishing APIs and webhooks into the platform.

Engineer Jan 2017 - May 2020

Outstart | Lahore, Pakistan

- Engineered a multi-tenant property management platform for New York-based owners, automating maintenance requests and apartment service workflows end to end.
- Delivered LePark, a Laravel-based parking management system synchronizing in real time with field parking hardware (IoT).
- Architected secure REST APIs bridging web management dashboards and on-site IoT parking infrastructure.

PROJECTS

Premion DSP — Real-Time Bidding Platform (Demand-Side)

Node.js, Redis, Snowflake, OpenRTB, CTV/OTT

- High-throughput real-time bidding engine processing OpenRTB bid requests for CTV/OTT inventory: audience targeting, dynamic bid valuation and automated campaign budget pacing.
- Architected low-latency Node.js and Redis microservices holding sub-100ms bid responses under ad-exchange SLA.
- Owned the big-data telemetry pipeline into Snowflake and coordinated cross-functional delivery against those SLAs.

Premion Fate System — Ad-Tech Reporting & Campaign Management

Vue.js, NestJS, PostgreSQL, Snowflake

- Enterprise ad-tech reporting and campaign management platform: high-volume analytics for OTT/CTV location, sale and web attribution, creative media uploads and inventory forecasting.
- Owned the system architecture and established the API documentation standards the squads worked to.
- Directed the frontend and backend engineering squads through structured Agile delivery.

SophieFlow — AI Social Media Management SaaS (Founder / Sole Engineer)

Laravel 12, Vue 3 + TypeScript, Inertia.js, Redis, Stripe, Google Gemini / OpenRouter

- Multi-tenant SaaS to schedule and publish content across 12+ social networks, with a unified inbox, analytics, an AI assistant and CRM-style lead capture.
- Built an async publishing pipeline with queued per-account jobs, retry/backoff and per-platform formatting; 10+ OAuth integrations; Stripe billing; and role- and plan-based access control.

OneParacon — Construction Management SaaS (Sole Engineer, Client Project)

Laravel 13, React 18 + TypeScript, PostgreSQL, PWA, Anthropic Claude API

- Multi-tenant platform with QR site sign-on (GPS geofencing and offline queue), site diaries, defects, a Gantt programme, finance and compliance reporting.
- Added an AI assistant via Claude tool-calling with role-scoped tools and human confirmation, plus WHS-compliant immutable, audit-logged records.

Resume Submitter — AI Job-Application Automation (Sole Engineer, Client Project)

Laravel 13, Vue 3, Inertia.js, Anthropic Claude API, Gmail API, Chrome Extension (MV3)

- Tailors resumes to job posts from a master profile, generates cover letters, and emails applications through Gmail with the PDF attached.
- Built a deterministic keyword-matching engine, ATS-readability checks, prompt-injection guards, and a Manifest V3 Chrome extension for capturing and auto-filling applications.

SKILLS

Skills: OpenRTB, Real-Time Bidding (RTB), Audience Targeting, Big Data Pipelines, Software Architecture, System Design, Cost Optimization, RFC Writing, NestJS, Next.js, GitHub Actions, Continuous Integration/Continuous Deployment (CI/CD), Test-Driven Development (TDD), Agile/Scrum, Extract, Transform, Load (ETL) Data Migration, OpenRouter, Automated QA

Languages: PHP, JavaScript (ES6+), TypeScript

Backend: Laravel, Node.js (Express, Fastify), Microservices

Frontend: Vue.js (Vue 3), React

Databases & Data: PostgreSQL, MySQL, MongoDB, Snowflake, Redis, Query Optimization

AI & LLM: Anthropic Claude API, Google Gemini

Cloud & DevOps: Amazon Web Services (AWS: EC2, RDS, S3, Load Balancing), Docker, Nginx, PM2

Leadership: Code Review & Approval, Technical Direction, Mentoring, Sprint Planning

Domain: Multi-Tenant SaaS

EDUCATION

University of Lahore

2013 - 2017

B.S. Computer Science